A Nonunimodal Independence Polynomial of a Very Well-Covered Graph

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Abstract. Levit and Mandrescu conjectured that every very well-covered graph has a unimodal independence polynomial. We disprove this statement with the whiskering of $E_{588} \vee (39K_{11} \sqcup 85K_{12})$, a graph on 4,074 vertices. Its independence polynomial has the compact exact form

$$(1+x)^{1449}(1+2x)^{588} + (1+x)^{1913}(1+12x)^{39}(1+13x)^{85} - (1+x)^{2037}.$$

Direct integer evaluation gives the strict local valley $a_{1094} > a_{1095} < a_{1096}$. Two full-sequence constructions, including a differential recurrence, reproduce the result exactly.

1 Result and Construction

All graphs are finite and simple. Let $i_k(G)$ count the independent k-subsets of $V(G)$, let $\alpha(G)$ be the maximum size of an independent set, and write

$$I(G;x) = \sum_{k=0}^{\alpha(G)} i_k(G)x^k, \qquad i_0(G) = 1,$$

for the independence polynomial of Gutman and Harary [5]. A sequence is *unimodal* if it is weakly increasing up to some index and weakly decreasing thereafter, and *log-concave* if $i_k^2 \geq i_{k-1}i_{k+1}$ for all internal k; a positive log-concave sequence is unimodal [8]. Independence sequences of arbitrary graphs can fail unimodality in essentially arbitrary ways [1]. A graph is *well-covered* [15] if all of its maximal independent sets have the same cardinality. It is *very well-covered* [4] if, in addition, it has no isolated vertices and $|V(G)| = 2\alpha(G)$.

In the conclusions of [11] (Conjecture 3.1 of the arXiv version, math/0406623), Levit and Mandrescu conjecture:

Conjecture (Levit–Mandrescu [9, 11])**.** *$I(G;x)$ is log-concave for any very well-covered graph G.*

The weaker unimodality form of the statement is how the conjecture is recorded and studied in later literature: "Levit and Mandrescu conjectured that the independence polynomial of every very well-covered graph is unimodal" [2]. This paper refutes both forms.

Write E_s for the edgeless graph on s vertices, rK_m for the disjoint union of r copies of K_m, $A \vee B$ for graph join, and $W(F)$ for the graph obtained by attaching one private leaf v' to every $v \in V(F)$.

Theorem 1. *The graph*

$$H = W(F), \qquad F = E_{588} \vee (39K_{11} \sqcup 85K_{12}),$$

is very well-covered, and $I(H;x)$ is not unimodal.

Here is a deterministic edge rule for F. Its vertices are

$$A = \{a_i : 1 \leq i \leq 588\},$$
$$B = \{b_{r\ell} : 1 \leq r \leq 39,\ 1 \leq \ell \leq 11\},$$
$$C = \{c_{s\ell} : 1 \leq s \leq 85,\ 1 \leq \ell \leq 12\}.$$

Each fixed-r set of b-vertices and fixed-s set of c-vertices is a clique. Every vertex in A is adjacent to every vertex in $B \cup C$; there are no other base edges. The graph H adds v' and the single edge vv' for each $v \in V(F)$. Thus $|V(F)| = 588 + 39(11) + 85(12) = 2037$.

Let M be a maximal independent set of H. For every old vertex v, at most one of v, v' lies in M. At least one lies in M, because otherwise the private leaf v', whose only neighbor is v, could be added. Thus M contains exactly one vertex from each of the 2037 pairs $\{v, v'\}$. Every maximal independent set has size 2037, so $\alpha(H) = 2037$. The graph has no isolated vertices and $|V(H)| = 4074 = 2\alpha(H)$, proving that it is very well-covered.

2 Polynomial Certificate

Independent sets in a join lie wholly on one side, and independence polynomials multiply over disjoint union. The empty set is counted on both sides of the join, so

$$I(F;x) = (1+x)^{588} + (1+11x)^{39}(1+12x)^{85} - 1. \quad (1)$$

For an arbitrary n-vertex graph Q, fix the chosen old vertices $S \subseteq V(Q)$ before choosing leaves. If S is independent, precisely the $n - |S|$ leaves over unchosen vertices remain available. Hence

$$I(W(Q);x) = \sum_{S \text{ independent in } Q} x^{|S|}(1+x)^{n-|S|}$$
$$= (1+x)^n I\left(Q; \frac{x}{1+x}\right). \quad (2)$$

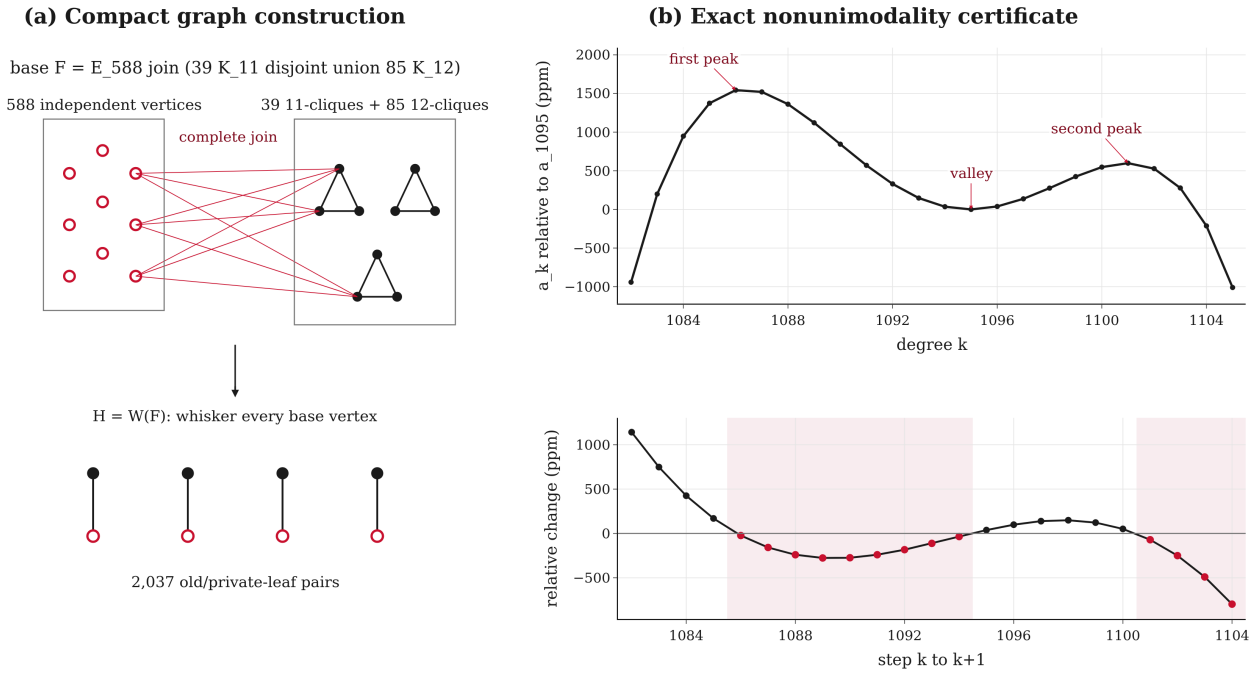


Figure 1: The compact construction has two peaks separated by an exact local valley. (a) Displayed base vertices and cliques are representative; the labels and the edge rule in the text specify the full graph. (b) The upper profile normalizes exact coefficients by a_{1095} only for display. The lower panel plots $10^6(a_{k+1}-a_k)/a_k$; red points are decreases.

This is also the K_1-corona specialization of Zhu's clique-cover-product identity [17]. Substituting (1) into (2) gives the complete exact polynomial

$$\begin{aligned} I(H;x) &= (1+x)^{1449}(1+2x)^{588} \\ &\quad + (1+x)^{1913}(1+12x)^{39}(1+13x)^{85} \\ &\quad - (1+x)^{2037}. \end{aligned} \tag{3}$$

Let $a_k = [x^k]I(H;x)$, with $\binom{u}{v} = 0$ for $v \notin \{0,\dots,u\}$. The Cauchy products in (3) give the directly evaluable formula

$$\begin{aligned} a_k &= \sum_{i=0}^{588} \binom{588}{i} 2^i \binom{1449}{k-i} - \binom{2037}{k} \\ &\quad + \sum_{i=0}^{39}\sum_{j=0}^{85} \binom{39}{i} 12^i \binom{85}{j} 13^j \binom{1913}{k-i-j}. \end{aligned} \tag{4}$$

Exact evaluation of (4) gives the two positive integers in Figure 2. Therefore $a_{1094} > a_{1095} < a_{1096}$. A weakly unimodal sequence cannot have a strict local minimum, which completes the proof of Theorem 1.

3 Independent Verification

The certificate uses no floating-point comparison. Equation (4) evaluates the three local coefficients directly. A separate construction obtains all 2038 coefficients by repeated exact multiplication: starting from [1], multiplication by $1+sx$ updates the coefficient array backward by $q_{k+1} \leftarrow q_{k+1} + sq_k$. It builds the three terms of (3) from slope multisets

$$\{1^{1449}, 2^{588}\}, \quad \{1^{1913}, 12^{39}, 13^{85}\}, \quad \{1^{2037}\}.$$

The direct sums and full construction agree at every certified local index.

A third exact construction follows from logarithmic differentiation. For $Q(x) = \prod_{h=1}^{d}(1+s_hx)^{e_h}$, set

$$D(x) = \prod_h (1+s_hx), \qquad R(x) = \sum_h e_h s_h \prod_{\ell \neq h}(1+s_\ell x).$$

Logarithmic differentiation gives the polynomial identity $D(x)Q'(x) = R(x)Q(x)$. Writing $D = \sum d_j x^j$, $R = \sum r_j x^j$, and $Q = \sum q_k x^k$, comparison of x^k determines q_{k+1} exactly:

$$(k+1)q_{k+1} = \sum_j r_j q_{k-j} - \sum_{j\geq 1} d_j (k-j+1) q_{k-j+1}, \tag{5}$$

where unavailable indices contribute zero. Every division by $k+1$ has zero remainder. Applying (5) separately to the three products in (3) reproduces all coefficients and the direct gaps.

Define $\Delta_k = a_{k+1} - a_k$. Both full-sequence methods give

```
D_minus = a_1094 - a_1095 =
412099709723481911757955943983840451737195637140184248625334380669639971541465402679742706117307459222023597297023042087
581608922521165299824351741747012699348576860335105740598589462754803295610432133099709828154536464659770654959846444372
246497083321366094784845792828381231336837486824251294380790483113948422439996384117565124895570802603732766419881557373
232708800809339561105966723997413007712645642511984771573436330096420576012314444567688150661819364169338847491458910590
622181307978205680436184942043773968947503751167634518221967864949838081554685495122082790103651367471414949279369668587
638054401666122291927686681705992845077006896464664441040067336715728842336211631704601676221586671557196413187

D_plus = a_1096 - a_1095 =
445830056393012364904188902031826586006858806246144804209531741717496460888290195422238653301546207227383742018965568282
392417006176126519475648194440900611274134166310725938467748831050455953892777799361816434039504789268323057421007344599
142829500231677878925185357012642122895124542523173836438029571674127153938105645010806829501036679753004935936562361287
446431391383006983006166467787899226817561947048937031638266200660487761699068149498264120397359535000494369021921382880
236124460566839256532596241499576905735124350387646175739246797056719849976320721232370820294437089819665437179778944517
526744843491529892237617731286722953291203640570438304286061110605486644731059008916049985656540975164476589187
```

Figure 2: Both exact adjacent differences are positive. The displayed differences are obtained by direct integer evaluation of (4); only line breaks are reflowed. Each difference has 711 decimal digits.

sign of Δ_k	inclusive range of k
$+$	0–1085
$-$	1086–1094
$+$	1095–1100
$-$	1101–2036

There are no zero differences. Endpoint checks are

$$a_0 = 1, \quad a_1 = 4074, \quad a_{2037} = 2^{588} + 12^{39}13^{85} - 1.$$

For an integrity check, serialize the full sequence as base-ten ASCII integers separated by single commas, with no whitespace, trailing comma, or newline. Its SHA-256 is

```
2fa9b0d7d14adef8d3070acf71dce163a89a7033571c65e49f7187131ccfa77a
```

4 Structure and Discovery

The search considered $W(A \vee B)$ with A and B disjoint unions of cliques. If their clique sizes are $a_1, \ldots, a_p$ and $b_1, \ldots, b_q$, respectively, then (2) yields

$$\begin{aligned} I(W(A \vee B); x) = (1+x)^{n-p} \prod_i (1 + (a_i + 1)x) \\ + (1+x)^{n-q} \prod_j (1 + (b_j + 1)x) \\ - (1+x)^n. \end{aligned}$$

At $x = 1$, each positive summand is a scaled Poisson-binomial profile. For the two terms in (3), their total masses M_A, M_B satisfy

$$\frac{M_A}{M_B} = \frac{3^{588}}{2^{549}13^{39}7^{85}} = 1.01089\ldots,$$

while their normalized coefficient distributions have exact means $2233/2$ and $7500/7$, separated by $631/14$. Thus the two components have almost equal mass but centers separated by about two standard deviations; their overlap creates the two peaks in Figure 1. The subtracted $(1+x)^{2037}$ term has total mass smaller than the first positive term by the factor $(3/2)^{588}$. A two-clique-size histogram screen used these mass and moment formulas to propose the displayed parameters; exact arithmetic alone promoted the hit. This mechanism explains the construction but proves neither the signs nor minimality. No claim that 4074 is the smallest possible order is made.

Appendix A gives a standalone exact computation of the entire coefficient sequence, its sign runs, its hash, and the two local gaps.

5 Context

Brown, Dilcher, and Nowakowski conjectured unimodality for all well-covered graphs in 2000 [3]. Michael and Traves refuted that parent statement for $\alpha \in \{4, 5, 6, 7\}$ and posed the roller-coaster conjecture [13]; Levit and Mandrescu extended the refutation to every $\alpha \geq 8$, noting explicitly that none of those counterexamples is very well-covered and that the conjecture remained open in that subclass [10]. For the very-well-covered case they conjectured the stronger log-concavity property quoted in Section 1 [9, 11]. The unimodality form is the statement attributed to them and studied since [2]; it was still reported open in March 2026 [12]. Theorem 1 refutes both formulations.

Positive partial results bracket the present construction. Levit and Mandrescu proved unimodality when $\alpha \leq 9$, log-concavity when $\alpha \leq 5$, and, in general, weak nondecrease through $\lceil \alpha/2 \rceil$ and weak nonincrease from $\lceil (2\alpha-1)/3 \rceil$ (Theorem 2.5 of the arXiv version of [11]). Here $\alpha(H) = 2037$, so those ranges end at 1019 and begin at 1358; the valley at 1095 lies in the previously uncontrolled interval. Brown and Cameron proved that every graph embeds as an induced subgraph of a very well-covered graph with unimodal independence polynomial, and recorded the leafy-extension range $\alpha(F) \leq 8$ [2]; the base here has $\alpha(F) = 588$. Zhu's preservation theorem for clique cover products [17] does not apply either: the base polynomial (1) is itself

non-log-concave, with first-difference sign pattern $+$ (0–114), $-$ (115–124), $+$ (125–293), $-$ (294–587). Recent sufficient conditions for log-concavity of $\mathbf{W}_p$ graphs [6, 14] also do not cover this graph: their thresholds require either $p \gtrsim \alpha^2/(4(\alpha+1))$ or order at least 3α, while a very well-covered graph is a $\mathbf{W}_1$ graph of order exactly 2α.

The graph contains large cliques, so this construction does not address the separate unimodality conjecture for trees [1]. In that neighboring problem, log-concavity fails starting at order 26 [7], machine search has since produced tens of thousands of further non-log-concave trees [16], yet all known examples remain unimodal [12].

A Executable Certificate

The following standalone Python program uses only exact integer arithmetic and the standard library. It constructs all coefficients from (3), checks the endpoints, positivity, every first-difference sign run, and the full-sequence hash, and prints the two strict local gaps. No external data, project file, or third-party package is required.

```
#!/usr/bin/env python3
"""Minimal exact certificate for the counterexample."""

from hashlib import sha256


def product(factors):
    coefficients = [1]
    for slope, exponent in factors:
        for _ in range(exponent):
            coefficients.append(slope * coefficients[-1])
            for k in range(len(coefficients) - 2, 0, -1):
                coefficients[k] += slope * coefficients[k - 1]
    return coefficients


def check(condition):
    if not condition:
        raise RuntimeError("verification failed")


first = product(((1, 1449), (2, 588)))
second = product(((1, 1913), (12, 39), (13, 85)))
baseline = product(((1, 2037),))
a = [u + v - w for u, v, w in zip(first, second, baseline)]
check(len(first) == len(second) == len(baseline) == 2038)
check(a[0] == 1 and a[1] == 4074)
check(a[-1] == 2**588 + 12**39 * 13**85 - 1)
check(all(value > 0 for value in a))
sign = [(v > u) - (v < u) for u, v in zip(a, a[1:])]
runs = []
for k, value in enumerate(sign):
    if not runs or runs[-1][0] != value:
        runs.append([value, k, k])
    else:
        runs[-1][2] = k
check(runs == [[1, 0, 1085], [-1, 1086, 1094],
               [1, 1095, 1100], [-1, 1101, 2036]])
check(a[1094] > a[1095] < a[1096])
digest = sha256(",".join(map(str, a)).encode("ascii")).hexdigest()
expected = (
    "2fa9b0d7d14adef8d3070acf71dce163"
    "a89a7033571c65e49f7187131ccfa77a"
)
check(digest == expected)
print("coefficient SHA-256 =", digest)
print("a_1094-a_1095 =", a[1094] - a[1095])
print("a_1096-a_1095 =", a[1096] - a[1095])
```

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